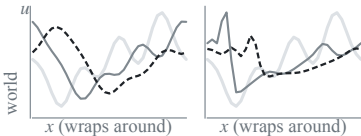


(a) A wave on a ring

temporal stability

seen: — first — last - - - true future



advection-diffusion
drifts and spreads

Burgers
steepens into a front

8 frames of the profile $u \rightarrow$ next 8

one denoiser, two samplers

predicted frames, in time order $\longrightarrow$

joint denoising

all partly noisy



causal freezing

still pure noise

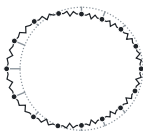


8 network calls each, shown after two

(b) Beads on a spring ring

spatial locality

..... rest position -•- offset from rest



linear or cubic springs

each bead is tied to its two neighbours

8 frames of offsets, velocities $\rightarrow$ next 8

which beads exchange messages



true ring
neighbours



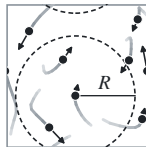
permuted ring
random partners



all pairs
no locality

(c) Repelling particles

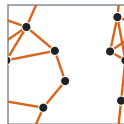
spatial locality



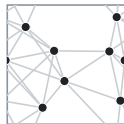
8 balls in a box whose edges wrap around
push apart when closer than R

8 frames of positions, velocities $\rightarrow$ next 8

which balls exchange messages



true radius
closer than R only



all pairs
any distance